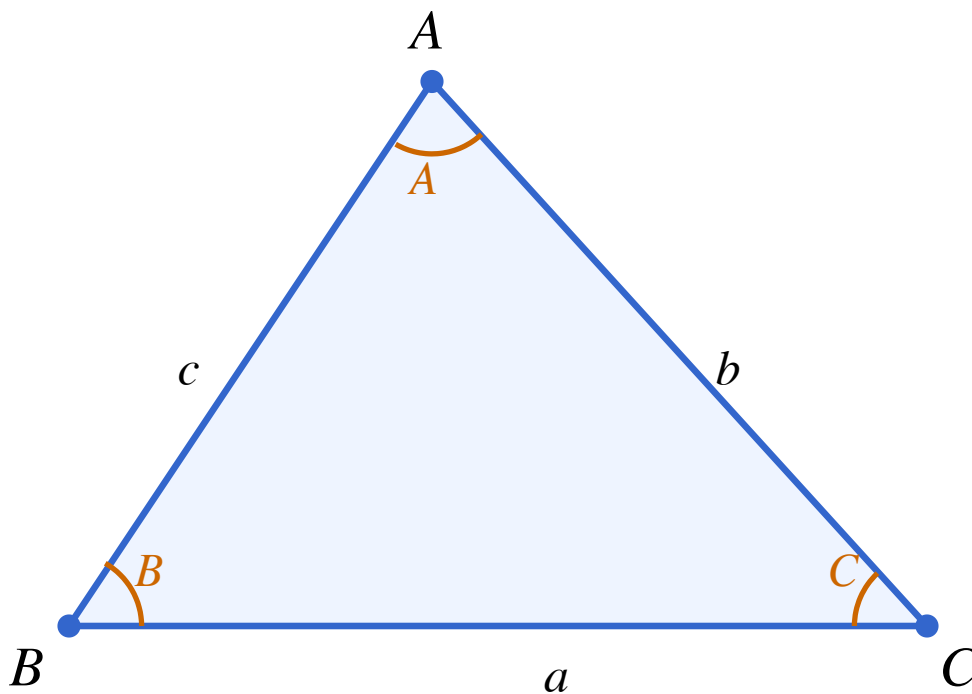


Lecture 16: Triangle Laws

Throughout this lecture, $\triangle ABC$ has sides $a = BC$, $b = CA$, $c = AB$ opposite to angles A , B , C respectively, circumradius R , inradius r , and semiperimeter $s = \frac{a + b + c}{2}$.

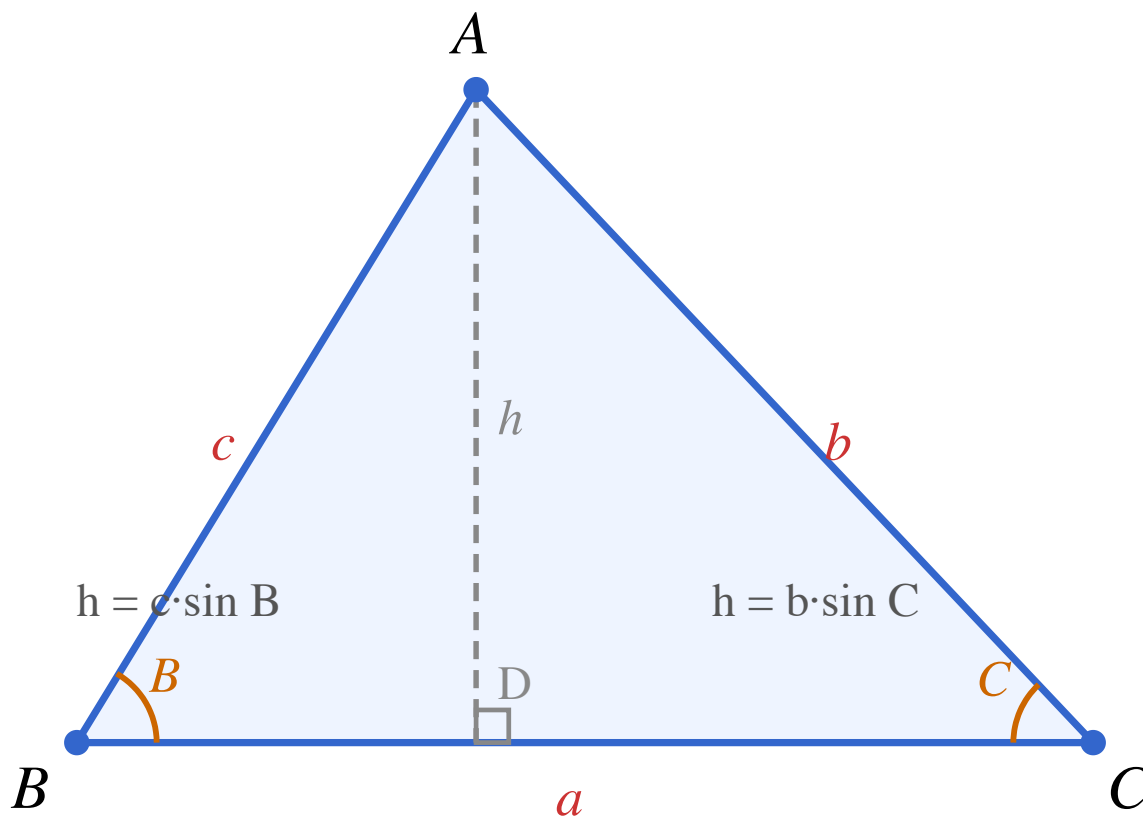


$$a=BC, b=CA, c=AB$$

Part 1: The Sine Law

Theorem 1 (Sine Law): In any triangle ABC ,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$



$$\frac{b}{\sin B} = \frac{c}{\sin C} = \frac{a}{\sin A}$$

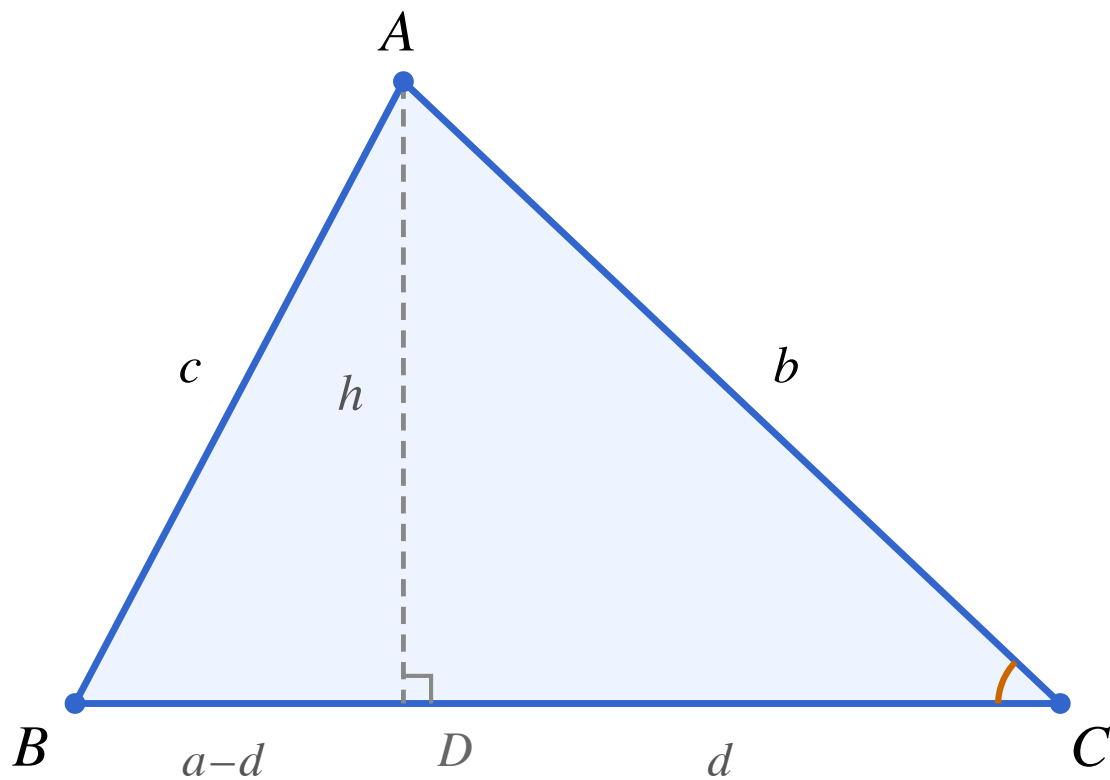
Proof: Draw the altitude from A to BC with foot D , and let $h = AD$. In right triangle ABD , $\sin B = h/c$; in right triangle ACD , $\sin C = h/b$. Hence $c \sin B = h = b \sin C$, giving $\frac{b}{\sin B} = \frac{c}{\sin C}$. By symmetry, repeating with altitudes from B and C yields all three ratios equal. ■

Part 2: The Cosine Law

Theorem 2 (Cosine Law): In any triangle ABC ,

$$c^2 = a^2 + b^2 - 2ab \cos C.$$

Similarly, $a^2 = b^2 + c^2 - 2bc \cos A$ and $b^2 = a^2 + c^2 - 2ac \cos B$.



Proof: Draw the altitude from A to BC with foot D , and let $h = AD$. In right triangle ACD : $d = b \cos C$ and $h^2 = b^2 - d^2$. Since $BD = a - d$, the Pythagorean theorem on $\triangle ABD$ gives

$$c^2 = h^2 + (a - d)^2 = (b^2 - d^2) + (a^2 - 2ad + d^2) = a^2 + b^2 - 2ad.$$

Substituting $d = b \cos C$ yields $c^2 = a^2 + b^2 - 2ab \cos C$. ■

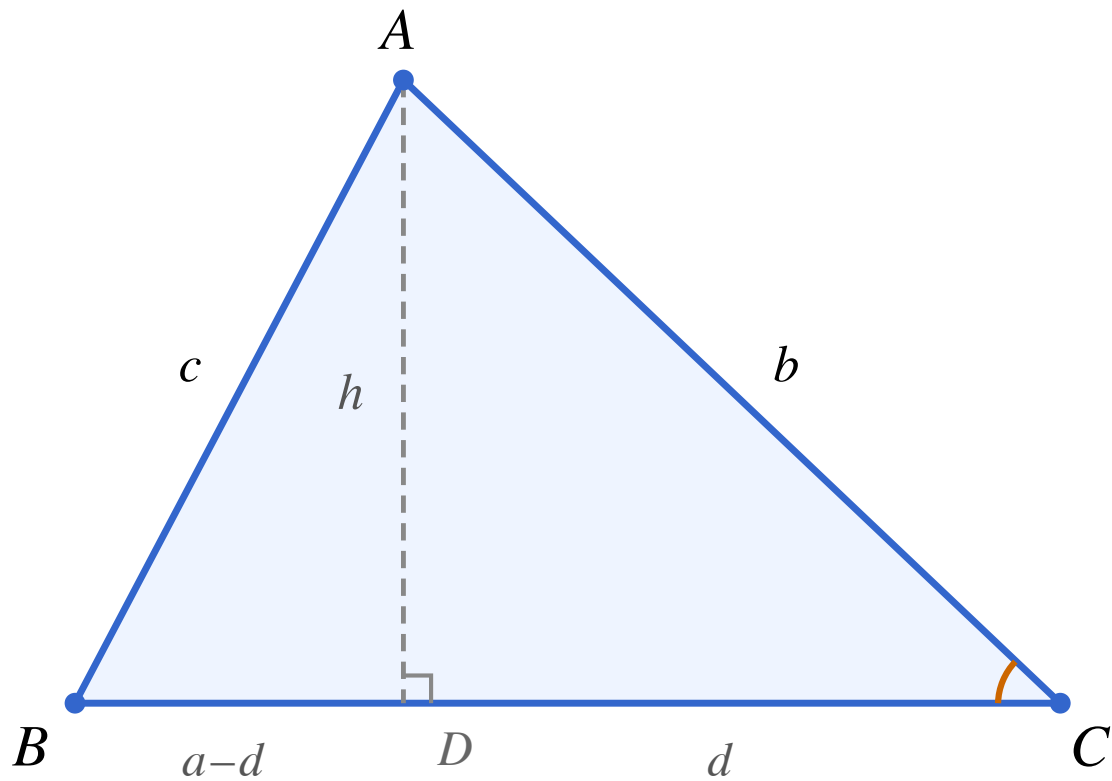
Remark: When $C = 90^\circ$, $\cos C = 0$ and the cosine law reduces to the Pythagorean theorem $c^2 = a^2 + b^2$.

Corollary: $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$, which determines C uniquely given a, b, c .

Part 3: Heron's Formula

Theorem 3 (Heron's Formula): The area of a triangle with sides a, b, c and semiperimeter $s = \frac{a+b+c}{2}$ is

$$S = \sqrt{s(s-a)(s-b)(s-c)}.$$



Proof: Use the same altitude h from A to foot D on BC . From the cosine law proof, $d = CD = \frac{a^2 + b^2 - c^2}{2a}$ and $h = \sqrt{b^2 - d^2}$. The area is $S = \frac{1}{2}ah$, so:

$$\begin{aligned}
S &= \frac{1}{2}a\sqrt{b^2 - \left(\frac{a^2 + b^2 - c^2}{2a}\right)^2} \\
&= \frac{1}{2}\sqrt{a^2b^2 - \left(\frac{a^2 + b^2 - c^2}{2}\right)^2} \\
&= \frac{1}{4}\sqrt{4a^2b^2 - (a^2 + b^2 - c^2)^2} \\
&= \frac{1}{4}\sqrt{(2ab + a^2 + b^2 - c^2)(2ab - a^2 - b^2 + c^2)} \\
&= \frac{1}{4}\sqrt{((a + b)^2 - c^2)(c^2 - (a - b)^2)} \\
&= \frac{1}{4}\sqrt{(a + b + c)(a + b - c)(c + a - b)(c - a + b)} \\
&= \frac{1}{4}\sqrt{2s \cdot 2(s - c) \cdot 2(s - b) \cdot 2(s - a)} \\
&= \sqrt{s(s - a)(s - b)(s - c)}. \quad \blacksquare
\end{aligned}$$